Day 4 — Student Tasks

Part A — Objects

1. Create an object named `car` with these properties: `brand`, `model`, `year`, and `color`. Then print the whole object.
2. Using the `car` object, print the brand using dot notation and print the year using bracket notation.
3. Change the color of the car to `"black"`, then add a new property `price` with any number value. Print the object again.
4. Delete the `year` property from the car object, then print `car.year` and explain what you see in the console.
5. Create an object with this property key: `"student-name"` and value `"Sara"`. Also add `age: 20`. Print the student name correctly.
6. Create a nested object for a book: `title`, and `author` as an object with `firstName` and `lastName`. Print the author's last name.
7. For this object: `{ name: "Ali", age: 25, city: "Cairo" }`, print all keys using `Object.keys`, all values using `Object.values`, and check if it has a property named `"job"` using `hasOwnProperty`.
8. Create an object `settings` with `theme: "dark"` and `lang: "en"`. Use `Object.freeze` on it, then try to change `theme` to `"light"` and try to add `fontSize: 16`. Print the object and write what happened.

Part B — Arrays Basics

9. Create an array of 5 city names. Print the array and print its length.
10. Print the first city, the second city, and the last city in the array.
11. Add a city at the end using `push`, then add a city at the start using `unshift`. Print the array after each step.
12. Remove the last city using `pop`, then remove the first city using `shift`. Print the array after each step.
13. Using this array: `["HTML", "CSS", "JS", "React"]`, print the index of `"JS"`, check if `"Python"` exists using `includes`, and print the result.

14. Print all items in this array using `forEach`: `["pen", "book", "bag"]`. Also print each item with its index.
15. Using `for...of` on `["red", "green", "blue", "yellow"]`, print the colors one by one, but stop when you reach `"blue"` (do not print blue).
16. Start with `["A", "B", "C"]`. Add `"D"` and `"E"` at the end, remove the first item, then print the final array and its length.

Part C — Array Methods

17. Using `map`, convert this array to uppercase: `["apple", "banana", "cherry"]`. Print the new array and also print the original array.
18. Using `filter`, keep only the numbers greater than 50 from: `[10, 55, 30, 80, 45, 90]`.
19. From `["Cairo", "Giza", "Alex", "Aswan"]`, use `find` to get the first city that starts with `"A"`, and use `findIndex` to get its index.
20. Using `slice`, copy part of this array: `["a", "b", "c", "d", "e"]` from index 1 to index 4 (not including 4). Print the copy and the original.
21. Using `splice`, remove 2 items starting from index 1 in this array: `["one", "two", "three", "four", "five"]`. Print the removed items and the array after the change.
22. Sort this numbers array correctly from smallest to largest: `[40, 100, 1, 5, 25]`. Print the result.
23. Use `some` to check if any age in `[16, 21, 17, 19]` is greater than or equal to 18. Use `every` to check if all ages are greater than or equal to 18. Print both results.
24. Use `reduce` to calculate the total of: `[5, 10, 15, 20]`. Print the total.

Part D — Mix (Objects + Arrays)

25. You have this array of students: `[{ name: "Omar", grade: 80 }, { name: "Mona", grade: 90 }, { name: "Ali", grade: 70 }]` Use a loop to print each student name and grade.
26. From the same students array, use `filter` to get only students with grade greater than or equal to 80, then print their names only using `map`.
27. Create an array of product objects. Each product has `name` and `price`. Use `reduce` to calculate the total price of all products and print it.

28. You have: `["js", "html", "css", "js", "react", "js"]`. Print how many times `"js"` appears in the array.
29. Create an object `classroom` with: `teacher` (string), and `students` (array of 4 names). Print the teacher name, print the number of students, and print the last student name.
30. You have this array: `[{ id: 1, title: "Pen", price: 10 }, { id: 2, title: "Book", price: 50 }, { id: 3, title: "Bag", price: 25 }]`. Create a new array that contains only the titles in uppercase, then create another array that contains only products with price less than 30, and finally print the total of all prices.